12TH CHEMISTRY · CHAPTER 24 · FULL CHAPTER NOTES

Organic Nitrogen Compounds

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION

Q.1 [Short] Why are organic nitrogen compounds important?

Organic nitrogen compounds are essential to life and the modern world. They form the basis of life in the form of amino acids and nucleotides, making the most vital compounds for life, i.e. proteins. Amines and nitro compounds have pivotal significance in the dye industry and medicines. Nylon, Kevlar and polyurethane are nitrogen-containing polymers found in every home and office. Nitrogen compounds are also part of our food, fertilisers, explosives, etc.

Q.2 [Short] What is the difference between caffeine and theobromine?

Caffeine in coffee and theobromine in chocolate are both organic nitrogen compounds; they are the active compounds that make our mood happy. Their ring structures are the same. The difference is that caffeine has three methyl groups attached to nitrogen atoms, while in theobromine one of these N–CH_3 groups is replaced by N–H (it has only two methyl groups).

24.1 AMINES

Q.3 [Long] What are amines? Classify them into primary, secondary and tertiary amines with examples.

Amines are considered as organic derivatives of ammonia, in which alkyl or aryl groups replace one, two or three hydrogen atoms of ammonia. The number of hydrogen atoms replaced gives the classification into primary, secondary and tertiary amines. In the IUPAC system amines are named as alkanamines.

Type	Structure	Examples
Primary amine	One alkyl or aryl group attached to the $–NH_2$ group ($R–NH_2$)	methanamine, ethanamine, propan-1-amine, phenylamine (aniline)
Secondary amine	Two alkyl or aryl groups attached to the NH group (R_2NH)	N-methylmethanamine, N-methylethanamine, diphenylamine
Tertiary amine	Three alkyl or aryl groups attached to the same nitrogen atom (R_3N)	N,N-dimethylmethanamine, triphenylamine

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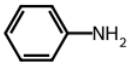
ethanamine (primary)



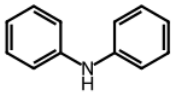
N-methylethanamine (secondary)



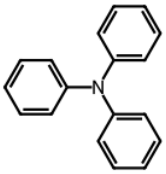
N,N-dimethylmethanamine (tertiary)



phenylamine (primary)



diphenylamine (secondary)



triphenylamine (tertiary)

Q.4 [Short] What are quaternary ammonium salts?

Tetra-substituted nitrogen derivatives are also known; they are called **quaternary ammonium salts**, e.g. tetramethylammonium chloride, $[(CH_3)_4N]^+Cl^-$.

Q.5 [Short] How does the classification of amines differ from that of alcohols and alkyl halides?

It is the branching at the **nitrogen atom**, not at the carbon atom attached to it, that determines whether an amine is primary, secondary or tertiary. This is not the same as in alcohols or monohaloalkanes (alkyl halides), where the carbon atom carrying the group is considered.

Quick Check 24.1

Q.6 [Short] a) Give IUPAC names of the following compounds. i) CH_3–$CH(CH_3)$–CH_2–CH_2–CH_2–NH_2 ii) CH_3–CH_2–$CH(NH)$–CH_3 iii) CH_3–CH_2–$CH(CH_2$–$CH_3)$–$CH(CH_2$–$CH_3)$–NH–CH_2–CH_3

i) 4-methylpentan-1-amine

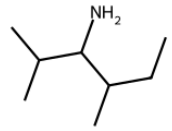
ii) butan-2-amine

iii) N,4-diethylhexan-3-amine (a secondary amine; the longest chain through the carbon bonded to nitrogen has six carbon atoms)

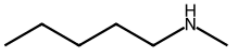
In part ii) the group is printed as "NH", and the textbook itself adds "(note: likely intended to be NH_2 for 2-butanamine)". It is named here as butan-2-amine.

Q.7 [Short] **b) Draw a structure corresponding to each amine. i) 2,4 dimethylhexan -3- amine ii) N-methylpentan -1- amine iii) 2-methylcyclohexan-1-amine iv) N, N -dimethylethanamine**

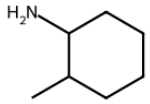
	Name	Condensed structure
i)	2,4-dimethylhexan-3-amine	CH_3–$CH(CH_3)$–$CH(NH_2)$–$CH(CH_3)$–CH_2–CH_3
ii)	N-methylpentan-1-amine	CH_3–CH_2–CH_2–CH_2–CH_2–NH–CH_3
iii)	2-methylcyclohexan-1-amine	cyclohexane ring with –NH_2 on C-1 and –CH_3 on C-2
iv)	N,N-dimethylethanamine	CH_3–CH_2–$N(CH_3)_2$



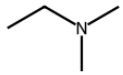
i) 2,4-dimethylhexan-3-amine



ii) N-methylpentan-1-amine



iii) 2-methylcyclohexan-1-amine



iv) N,N-dimethylethanamine

24.2 PREPARATION OF AMINES

Q.8 [Long] **Describe the methods of preparation of aliphatic primary amines.**

i) From halogenoalkanes and ammonia: primary amines are prepared by the nucleophilic substitution of halogenoalkanes with ammonia. Excess of ethanolic ammonia under pressure is used to avoid the formation of secondary amines or a mixture of products.

$$CH_3–CH_2–Br + NH_3 \text{ (excess)} \rightarrow CH_3–CH_2–NH_2 + HBr \text{ (ethanol, high pressure)}$$

ii) Reduction of alkanenitriles with H_2/Ni or $LiAlH_4$:

$$CH_3–CH_2–C{\equiv}N + 4[H] \rightarrow CH_3–CH_2–CH_2–NH_2 \ (LiAlH_4)$$

iii) Reduction of nitroalkanes in the presence of $LiAlH_4$:

$$CH_3–CH_2–NO_2 + 3H_2 \rightarrow CH_3–CH_2–NH_2 + 2H_2O \ (LiAlH_4)$$

iv) Reduction of alkanamides by $LiAlH_4$; the yield is good since no by-product is formed:

$$CH_3–CO–NH_2 + 4[H] \rightarrow CH_3–CH_2–NH_2 + H_2O \ (LiAlH_4;\ H_2O/H^+)$$

v) Hofmann reaction: this reaction yields the amine with one carbon less than the starting amide.

$$CH_3–CO–NH_2 + Br_2 + 4NaOH \rightarrow CH_3–NH_2 + 2NaBr + Na_2CO_3 + 2H_2O$$

Q.9 [Short] Why is an excess of ethanolic ammonia used in the preparation of a primary amine from a halogenoalkane?

The primary amine formed is itself a nucleophile and can react further with the halogenoalkane to give secondary amines or a mixture of products. With ammonia in excess, the halogenoalkane reacts mainly with ammonia, so the primary amine is the major product.

If NH_3 is in excess, the major product is the primary amine; if the halogenoalkane is in excess, a tertiary amine is the major product.

Q.10 [Short] What is the Hofmann reaction? What is special about its product?

It is the reaction of an amide with bromine and sodium hydroxide to give a primary amine. The amine formed has one carbon atom less than the starting amide.

$$CH_3–CO–NH_2 + Br_2 + 4NaOH \rightarrow CH_3–NH_2 + 2NaBr + Na_2CO_3 + 2H_2O$$

Q.11 [Long] How are aliphatic secondary amines prepared?

i) Alkylation of primary amines: when a halogenoalkane reacts with a primary amine in the presence of ethanol under pressure, a secondary amine is formed.

$$CH_3–CH_2–Br + CH_3–NH_2 \rightarrow CH_3–CH_2–NH–CH_3 + HBr \text{ (ethanol)}$$

ii) Reduction of N-substituted amides:

$$CH_3–CO–NH–CH_3 + 4[H] \rightarrow CH_3–CH_2–NH–CH_3 + H_2O \ (H_2 \text{ with Ni/Pt, or } LiAlH_4)$$

The textbook writes this reduction as $CH_3–CO–NH–CH_3 + H_2 \rightarrow CH_3–CH_2–NH–CH_3$ (Ni/Pt), which is not balanced; it is balanced above with 4[H] and H_2O.

Q.12 [Long] Describe the methods of preparation of phenylamine.

i) Ammonolysis of chlorobenzene: ammonolysis is a reaction in which an ammonia molecule is broken down to react with another compound.

$$C_6H_5–Cl + 2NH_3 \rightarrow C_6H_5–NH_2 + NH_4Cl \ (Cu_2O, 250\ °C, 50\ atm)$$

ii) Reduction of nitrobenzene:

$$C_6H_5–NO_2 + 6[H] \rightarrow C_6H_5–NH_2 + 2H_2O \ (Sn + \text{conc. } HCl)$$

iii) Hofmann reaction of benzamide:

$$C_6H_5–CO–NH_2 + Br_2 + 4NaOH \rightarrow C_6H_5–NH_2 + 2NaBr + Na_2CO_3 + 2H_2O$$

The three methods are numbered i, iii and iv in the textbook (there is no ii).

Q.13 [Short] What is ammonolysis?

Ammonolysis is a reaction in which an ammonia molecule is broken down to react with another compound, e.g. chlorobenzene with ammonia gives phenylamine.

Quick Check 24.2

Q.14 [Short] **a) The "feel good factor" in chocolate has been identified as 2-phenylethanamine, $C_6H_5CH_2CH_2NH_2$. Suggest a synthesis of this compound from chloromethylbenzene, $C_6H_5CH_2Cl$.**

Step 1: heat chloromethylbenzene with KCN in ethanol; the chlorine is replaced by –CN and the chain becomes one carbon longer.

$$C_6H_5–CH_2–Cl + KCN \rightarrow C_6H_5–CH_2–CN + KCl \text{ (ethanol, heat)}$$

Step 2: reduce the nitrile with $LiAlH_4$ (or H_2/Ni).

$$C_6H_5–CH_2–CN + 4[H] \rightarrow C_6H_5–CH_2–CH_2–NH_2 \ (LiAlH_4)$$

Q.15 [Short] **b) What nitro compound, nitrile and amide are reduced to yield each of the following amine compound? I) For the primary amine: CH_3–CH(CH_3)–CH_2–NH_2 (determine the structures of nitro compound, nitrile, and amide that would produce this amine upon reduction) ii) For the secondary amine: CH_3–CH_2–CH_2–CH_2–NH–CH_3**

I) 2-Methylpropan-1-amine is obtained by the reduction of:

Class	Compound	Structure
Nitro compound	2-methyl-1-nitropropane	$(CH_3)_2CH–CH_2–NO_2$
Nitrile	2-methylpropanenitrile	$(CH_3)_2CH–C\equiv N$
Amide	2-methylpropanamide	$(CH_3)_2CH–CO–NH_2$

ii) N-Methylbutan-1-amine is a secondary amine. It is obtained by the reduction of an N-substituted amide: N-methylbutanamide, $CH_3–CH_2–CH_2–CO–NH–CH_3$ (or N-butylmethanamide, $H–CO–NH–CH_2–CH_2–CH_2–CH_3$). The reduction of a nitro compound or a nitrile gives only a primary amine (–NH_2), so neither can give this secondary amine directly.

"Nh_2" and "Ch_3" are printed in the formulae of this Quick Check; they are –NH_2 and –CH_3.

24.3 REACTIONS OF AMINES — Basicity

Q.16 [Long] **Explain the basicity of amines. On what does it depend?**

Amines which are soluble in water form weakly alkaline solutions, just as ammonia does, due to partial reaction with the solvent forming OH^- ions.

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^- \ (pK_b = 4.75)$$
$$CH_3–CH_2–NH_2 + H_2O \rightleftharpoons CH_3–CH_2–NH_3^+ + OH^- \ (pK_b = 3.29)$$
$$C_6H_5–NH_2 + H_2O \rightleftharpoons C_6H_5–NH_3^+ + OH^- \ (pK_b = 9.38)$$

The basicity of amines depends on the availability of the lone pair of electrons on the nitrogen atom. The larger the K_b (or the smaller the pK_b), the stronger the base.

Electron-donating alkyl groups attached to nitrogen increase the basicity, whereas electron-withdrawing groups decrease it. Basicity is also an indicator of the nucleophilic character of a compound.

Sr.No.	Amine	Formula	K_b	pK_b
1	Ammonia	NH_3	1.8×10^{-5}	4.75
2	Ethanamine	$CH_3–CH_2–NH_2$	5.1×10^{-4}	3.29
3	Phenylamine (aniline)	$C_6H_5–NH_2$	4.2×10^{-10}	9.38

In the equation for ethanamine the textbook prints $pK_b = 4.75$ (the value of ammonia); its own Table 24.1 gives 3.29 for ethanamine.

Q.17 [Long] Compare the basicity of aqueous ammonia, ethanamine and phenylamine.

Ethanamine is a stronger base than ammonia. The ethyl group is electron-donating; it pushes electron density towards the nitrogen atom, so the lone pair is more available to accept a proton. Likewise, alkyl amines in general are stronger bases than ammonia.

Ammonia is a stronger base than phenylamine. In phenylamine the lone pair of electrons on the nitrogen atom is delocalised over the benzene ring. The nitrogen atom is planar, with its lone pair in a p orbital, which overlaps with the delocalised π electrons of the benzene ring. The lone pair is therefore less available to accept a proton.

Basic strength: ethanamine > ammonia > phenylamine

Q.18 [Short] How do amines react with acids?

Because of their basic nature, ammonia and amines react with acids to form salts. The nitrogen atom donates its electron pair to form a dative bond with a proton.

$$NH_3 + HCl \rightarrow NH_4^+Cl^-$$

$$CH_3–CH_2–NH_2 + HCl \rightarrow CH_3–CH_2–NH_3^+Cl^-$$

Q.19 [Short] Why does lemon juice or vinegar reduce the smell of fish?

The foul smell of fish is due to N,N-dimethylmethanamine (trimethylamine). When lemon juice is sprinkled on fish, its acid (citric acid) reacts with the amine to form a salt, making it less smelly. The same happens when fish is processed with vinegar.

Q.20 [Long] How can primary, secondary and tertiary amines be distinguished with nitrous acid?

Primary, secondary and tertiary aliphatic amines are identified and distinguished by reacting them with nitric(III) acid (nitrous acid), prepared in situ.

$$NaNO_2 + HCl \rightarrow HNO_2 + NaCl$$

Primary amines form diazonium salts; the process is called diazotization. Nitrogen gas is produced (seen as bubbles).

$$CH_3–CH_2–NH_2 + HNO_2 + HCl \rightarrow CH_3–CH_2–N_2^+Cl^- + 2H_2O$$ (ethanediazonium chloride)

Secondary amines form N-nitrosoamines (a yellow oily liquid).

$$CH_3–CH_2–NH–CH_3 + HNO_2 \rightarrow CH_3–CH_2–N(CH_3)–NO + H_2O$$
(N-methyl-N-nitrosoethanamine)

Tertiary amines form soluble nitrite salts; there is no visible sign of reaction.

$$CH_3–CH_2–N(CH_3)_2 + HNO_2 \rightarrow [CH_3–CH_2–NH(CH_3)_2]^+NO_2^-$$
(N,N-dimethylethanammonium nitrite)

Amine	Product with nitrous acid	Observation
Primary	Diazonium salt	Nitrogen gas (bubbles)
Secondary	N-nitrosoamine	Yellow oily liquid
Tertiary	Soluble nitrite salt	No visible change

Q.21 [Short] What is diazotization?

It is the reaction of a primary amine with nitrous acid (prepared in situ from sodium nitrite and HCl) to form a diazonium salt.

Quick Check 24.3

Q.22 [Short] a) Phenylamine is about a million times less basic than cyclohexylamine. Give reason. (phenylamine $K_b = 4.2 \times 10^{-10}$; cyclohexylamine $K_b = 3.3 \times 10^{-4}$)

In phenylamine the lone pair on nitrogen is delocalised over the benzene ring (it is in a p orbital which overlaps with the π electrons of the ring), so it is much less available to accept a proton. In cyclohexylamine the ring is saturated: there is no delocalisation, and the alkyl ring is electron-donating, which makes the lone pair more available.

Q.23 [Short] b) Which of ethanamine and propan-2-amine is more basic? Explain.

Propan-2-amine is more basic. Basicity depends on the availability of the lone pair on nitrogen, and electron-donating alkyl groups increase it. In propan-2-amine the carbon bonded to nitrogen carries two methyl groups, so more electron density is pushed towards nitrogen than by the ethyl group of ethanamine.

Q.24 [Short] c) Show the reaction of propan-2-amine with HCl. Name the salt formed.

$$(CH_3)_2CH–NH_2 + HCl \rightarrow (CH_3)_2CH–NH_3^+Cl^-$$

The salt is propan-2-ammonium chloride (isopropylammonium chloride).

Q.25 [Short] **d) Which acid reacts with N,N-dimethylammine in fish when it is treated with vinegar? Write a chemical equation to show this reaction.**

Vinegar contains **ethanoic acid** (acetic acid). It reacts with the amine of fish, N,N-dimethylmethanamine (trimethylamine), to form a salt which is less smelly.

$$(CH_3)_3N + CH_3\text{–}COOH \rightarrow (CH_3)_3NH^+\ CH_3\text{–}COO^-$$

24.3.3 REACTIONS OF PHENYLAMINE

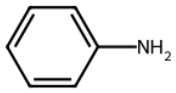
Q.26 [Short] **Which two types of reactions are given by phenylamine?**

Phenylamine gives two types of reactions: (a) due to the benzene ring, whose carbon atoms are sp^2 hybridized, and (b) due to the $–NH_2$ group, whose nitrogen is sp^3 hybridized.

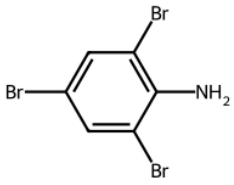
Q.27 [Long] **Describe the reaction of phenylamine with bromine water.**

The amino group ($–NH_2$) is an ortho-para directing group, so the incoming electrophile attacks the “o” and “p” positions. The amino group makes the benzene ring fairly reactive, especially at these positions. Hence bromination of phenylamine is faster than that of benzene, and its two ortho positions and the para position are all occupied in a single reaction. A white precipitate of 2,4,6-tribromophenylamine is formed.

$$C_6H_5\text{–}NH_2 + 3Br_2 \rightarrow C_6H_2Br_3\text{–}NH_2 + 3HBr$$



phenylamine



2,4,6-tribromophenylamine (white ppt)

Q.28 [Long] **How does phenylamine react with nitrous acid? Why is the mixture kept below 10 °C?**

Phenylamine, being a primary amine, reacts with nitrous acid to form a phenyldiazonium salt. This reaction is called **diazotization**. Organic compounds having the diazonium ion, $–N^+{\equiv}N$ (diazo group), as functional group are called diazo compounds. Nitrous acid is unstable and is prepared in situ from sodium nitrite and HCl. The reaction mixture is kept below 10 °C using ice, because diazonium salts are unstable and decompose easily, giving off nitrogen gas, even at room temperature.

$$NaNO_2 + HCl \rightarrow HNO_2 + NaCl$$

$$C_6H_5\text{–}NH_2 + HNO_2 + HCl \rightarrow C_6H_5\text{–}N_2^+Cl^- + 2H_2O \text{ (less than 10 °C)}$$

$$C_6H_5\text{–}N_2^+Cl^- + H_2O \rightarrow C_6H_5\text{–}OH + N_2 + HCl \text{ (heat)}$$

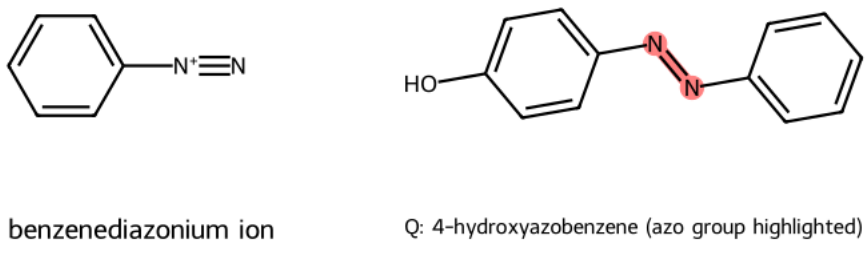
Q.29 [Short] Why are aryl diazonium salts more stable than alkyl diazonium salts?

Alkyl diazonium salts are highly unstable under all conditions and therefore cannot be isolated. Aryl diazonium salts are fairly stable at low temperatures. The extra stability of aromatic diazo compounds is due to the delocalisation of the positive charge over the aromatic ring. Benzenediazonium salts are important intermediates in the synthesis of many organic compounds.

Q.30 [Long] What are coupling reactions? Explain the formation of an azo compound with its mechanism.

The reactions of aryl diazonium salts with strongly activated aromatic compounds, such as phenols and amines, are called **coupling reactions**. The products, which contain an –N=N– bond, are called **azo compounds**. In the coupling reaction the positively charged diazonium ion acts as an electrophile and substitutes into the benzene ring of phenol at the para position.

$$C_6H_5–N_2^+Cl^- + C_6H_5–OH \rightarrow C_6H_5–N=N–C_6H_4–OH + HCl\ (NaOH)$$



P: benzenediazonium ion Q: 4-hydroxyazobenzene (azo group highlighted)

Benzenediazonium ion and p-hydroxyazobenzene (orange solid); the azo group is highlighted

Mechanism: the terminal nitrogen of the diazonium ion (the electrophile) is attacked by the electron-rich para carbon of phenol, giving an intermediate that carries a positive charge; this then loses H^+ from that carbon, the ring becomes aromatic again and p-hydroxyazobenzene is formed.

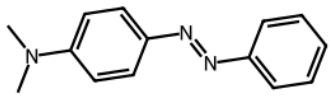
Q.31 [Short] Under what conditions does a diazonium salt couple with (a) phenol and (b) aromatic amines?

Coupling with phenol occurs in mild alkaline solution, and coupling with aromatic amines occurs in mild acidic conditions.

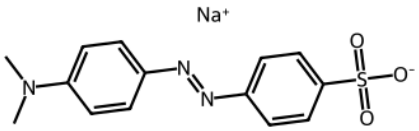
Q.32 [Long] How are azo dyes formed? Describe butter yellow and methyl orange.

Coupling reactions are used to form different types of azo dyes. **p-Dimethylaminoazobenzene** is called butter yellow; it was previously used in food, but because of its carcinogenic nature its use in food is now forbidden. It is made by coupling benzenediazonium chloride with N,N-dimethylphenylamine. **Methyl orange**, an acid–base indicator, is prepared from diazotized sulfanilic acid and N,N-dimethylphenylamine.

$$C_6H_5–N_2^+Cl^- + C_6H_5–N(CH_3)_2 \rightarrow C_6H_5–N=N–C_6H_4–N(CH_3)_2 + HCl\ (CH_3COOH)$$



p-dimethylaminoazobenzene (butter yellow)



methyl orange

Q.33 [Short] What is a chromophore?

The azo group (Ph–N=N–Ph) in azo dyes is called a **chromophore**. This group is responsible for the intense colour of the dyes, ranging from yellow and orange to red, blue and green, depending on what other groups are attached to the benzene rings. Different food colours often contain azo dyes.

Q.34 [Short] Why is butter yellow no longer used in food?

p-Dimethylaminoazobenzene (butter yellow) was previously used in food. It is carcinogenic (cancer-causing) in nature, so its use in food is now forbidden.

Quick Check 24.4

Q.35 [Short] Azo dyes are the coloured compounds which are prepared by the reaction of diazonium salt of phenylamine and some aryl compound having activated benzene ring. i) How will you prepare a dye from phenylamine and 2-methylphenol. ii) Give equations stepwise along with reaction conditions.

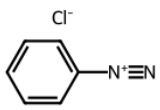
i) Phenylamine is first diazotized with nitrous acid below 10 °C. The cold benzenediazonium chloride solution is then added to 2-methylphenol dissolved in sodium hydroxide solution (mild alkaline conditions). The diazonium ion couples at the position para to the –OH group, giving an azo dye.

ii) Equations:

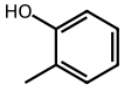
$$NaNO_2 + HCl \rightarrow HNO_2 + NaCl$$

$$C_6H_5–NH_2 + HNO_2 + HCl \rightarrow C_6H_5–N_2^+Cl^- + 2H_2O\ (\text{less than } 10\ °C)$$

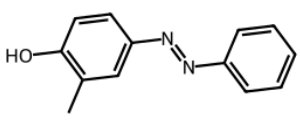
$$C_6H_5–N_2^+Cl^- + CH_3–C_6H_4–OH \rightarrow C_6H_5–N=N–C_6H_3(CH_3)–OH + HCl\ (NaOH)$$



benzenediazonium chloride



2-methylphenol



azo dye (4-hydroxy-3-methylazobenzene)

24.4 ALKANENITRILES

Q.36 [Long] What are alkanenitriles? Describe their acid and alkaline hydrolysis.

Organic compounds containing the –CN functional group are called alkanenitriles, e.g. ethanenitrile, CH_3–CN, and propanenitrile, CH_3–CH_2–CN. (Their preparation is given in Chapter 21.) Their most important application is their hydrolysis to carboxylic acids, which can occur with both dilute acid and dilute alkali.

Acid hydrolysis: when a nitrile is heated with a dilute acid (dilute HCl or dilute sulphuric acid), a carboxylic acid is formed.

$$CH_3CN_{(aq)} + 2H_2O_{(l)} + HCl_{(aq)} \rightarrow CH_3COOH_{(aq)} + NH_4Cl_{(aq)}$$

Alkaline hydrolysis: when a nitrile is heated with dilute sodium hydroxide solution, a salt of the carboxylic acid is produced, which on acidification with dilute HCl gives the free carboxylic acid.

$$CH_3CN_{(aq)} + H_2O_{(l)} + NaOH_{(aq)} \rightarrow CH_3COONa_{(aq)} + NH_{3(g)}$$
$$CH_3COONa_{(aq)} + HCl_{(aq)} \rightarrow CH_3COOH_{(aq)} + NaCl_{(aq)}$$

Q.37 [Short] How does the product of alkaline hydrolysis of a nitrile differ from that of acid hydrolysis?

Acid hydrolysis gives the carboxylic acid directly (with an ammonium salt). Alkaline hydrolysis gives the salt of the carboxylic acid and ammonia gas; the free acid is obtained only after acidification with dilute HCl.

24.5 AMIDES

Q.38 [Short] What are amides? Why are they called acid amides?

Amides contain the functional group –CO–NH_2. Since they are prepared from carboxylic acids, they are still sometimes called “acid amides”.

Q.39 [Long] Describe the methods of preparation of amides.

1. By the reaction of acyl chlorides with ammonia:

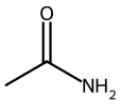
$$CH_3\text{–}COCl + NH_3 \rightarrow CH_3\text{–}CO\text{–}NH_2 + HCl$$
$$C_6H_5\text{–}COCl + NH_3 \rightarrow C_6H_5\text{–}CO\text{–}NH_2 \text{ (benzamide)} + HCl$$

2. By the reaction of acyl chlorides with primary amines: N-substituted amides are formed.

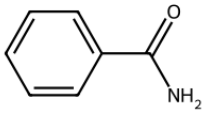
$$CH_3\text{–}COCl + CH_3\text{–}CH_2\text{–}NH_2 \rightarrow CH_3\text{–}CO\text{–}NH\text{–}CH_2\text{–}CH_3 \text{ (N-ethylethanamide)} + HCl$$
$$CH_3\text{–}COCl + C_6H_5\text{–}NH_2 \rightarrow CH_3\text{–}CO\text{–}NH\text{–}C_6H_5 \text{ (N-phenylethanamide)} + HCl$$

All these reactions take place at room temperature, releasing white fumes of hydrogen chloride. The HCl produced reacts with another molecule of ammonia or amine to form a salt; therefore an excess of ammonia or amine is used to ensure complete reaction.

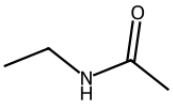
$$R\text{–}NH_2 + HCl \rightarrow R\text{–}NH_3^+Cl^-$$
$$NH_3 + HCl \rightarrow NH_4^+Cl^-$$



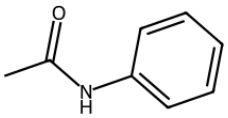
ethanamide



benzamide



N–ethylethanamide



N–phenylethanamide

Q.40 [Short] Why is an excess of ammonia or amine used in the preparation of amides from acyl chlorides?

The HCl produced in the reaction reacts with another molecule of ammonia or amine to form a salt. So an excess of ammonia or amine is used to ensure that the reaction is complete.

Q.41 [Short] Describe the physical properties of amides.

All amides except methanamide, which is a liquid, are solids with sharp melting points. They have quite high boiling points, as a consequence of strong intermolecular hydrogen bonding. In N-substituted amides, however, the melting and boiling points decrease, because the sites for hydrogen bonding diminish with an increasing number of alkyl substituents.

Q.42 [Long] Why are amides much weaker bases than amines?

Amides are less basic than aliphatic amines. Unlike amines, amides form neutral solutions in water. They exhibit resonance, in which the lone pair on nitrogen is delocalised with the oxygen of the carbonyl group. This reduces the availability of the electrons on nitrogen and so decreases the basicity.

The introduction of double-bond character between carbon and nitrogen reduces the C–N bond length to less than the usual C–N single bond length: 132 pm in an amide compared with 147 pm in an amine.

Q.43 [Short] Why are amides the least reactive of all acid derivatives?

Amides are the least reactive of all acid derivatives, because the lone pair of nitrogen is delocalised over the oxygen of the carbonyl group, which decreases the electrophilic power (δ+ charge) of the carbonyl carbon. Their reactions are generally slow and require prolonged heating or catalysts (acid or base).

Q.44 [Long] Describe the hydrolysis and reduction of amides.

1. Hydrolysis. With aqueous acid, unsubstituted amides yield the carboxylic acid and an ammonium salt. In the base-catalysed reaction the products are the salt of the carboxylic acid and ammonia; the free acid is obtained when this salt is treated with dilute hydrochloric acid.

$$CH_3–CO–NH_2 + H_2O + HCl \rightarrow CH_3–COOH + NH_4Cl$$
$$CH_3–CO–NH_2 + NaOH \rightarrow CH_3–COONa + NH_3$$
$$CH_3–COONa + HCl \rightarrow CH_3–COOH + NaCl$$

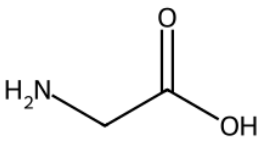
2. Reduction. Amides are reduced to primary amines with lithium tetrahydridoaluminate(III) in dry ethoxyethane (ether).

$$CH_3–CO–NH_2 + 4[H] \rightarrow CH_3–CH_2–NH_2 + H_2O\ (LiAlH_4, \text{dry ether})$$
$$C_6H_5–CO–NH_2 + 4[H] \rightarrow C_6H_5–CH_2–NH_2 + H_2O\ (LiAlH_4, \text{dry ether})$$

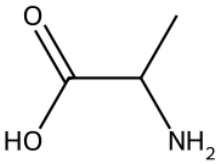
24.6 AMINO ACIDS

Q.45 [Long] What are amino acids? Why are they amphoteric? Give their general structure and examples.

Amino acids are an important group of compounds that contain the basic amino group ($–NH_2$) and the acidic carboxylic acid group (–COOH). This makes them **amphoteric**: they can behave as both an acid and a base. The general structure of amino acids (2-aminocarboxylic acids, also called α-amino acids) is $R–CH(NH_2)–COOH$. The R group is the part that varies in different amino acids. The simplest amino acid is glycine (aminoethanoic acid), in which R is a hydrogen atom; in alanine (2-aminopropanoic acid) R is the methyl group, $–CH_3$.



glycine (aminoethanoic acid)



alanine (2-aminopropanoic acid)

Q.46 [Short] What is a zwitterion? How is it formed?

An α-amino acid exists as a dipolar ion called a **zwitterion**, which has a positive and a negative end within the same molecule. It is formed when the proton goes from the carboxyl group to the amino group. The dipolar structure is also called the internal salt.

$$H_2N–CH(R)–COOH \rightarrow H_3N^+–CH(R)–COO^-$$

Q.47 [Short] What are essential and non-essential amino acids? Which amino acid is not chiral?

Except glycine, all α-amino acids are chiral. Out of 20 α-amino acids, 10 are essential (the body cannot synthesise them) and 10 are non-essential (the body can synthesise them).

Q.48 [Long] Explain the acidic and basic properties of amino acids.

Amino acids are amphoteric, but they are less acidic than most carboxylic acids and less basic than most amines. In fact, the acidic part of the amino acid molecule is the $-NH_3^+$ group, not a –COOH group, and the basic part is the $-COO^-$ group, not an $-NH_2$ group. RNH_3^+ is a weaker acid than RCOOH.

$$H_3N^+–CH(R)–COO^- + H_2O \rightleftharpoons H_2N–CH(R)–COO^- + H_3O^+ \ (pK_a \approx 10)$$
$$H_3N^+–CH(R)–COO^- + H_2O \rightleftharpoons H_3N^+–CH(R)–COOH + OH^- \ (pK_b \approx 12)$$

In aqueous solution an equilibrium exists between the dipolar ion and the cationic and anionic forms:

Form	Structure	Predominant at
Cationic form	$H_3N^+–CH(R)–COOH$	low pH (acidic)
Dipolar ion (zwitterion)	$H_3N^+–CH(R)–COO^-$	isoelectric pH
Anionic form	$H_2N–CH(R)–COO^-$	high pH (basic)

Q.49 [Short] What is the isoelectric point (pI) of an amino acid?

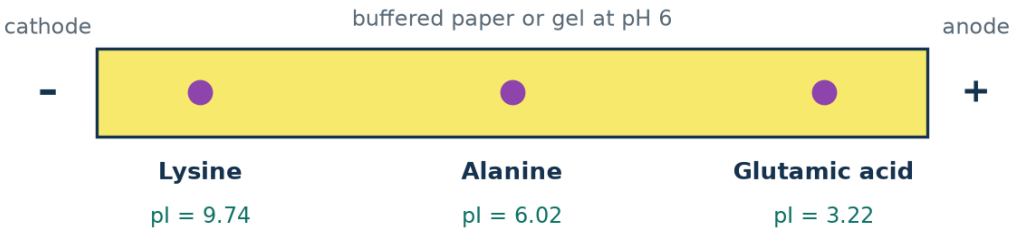
For each amino acid there is a specific pH at which the concentration of the zwitterion reaches its maximum value. This pH is called the **isoelectric point (pI)**; at it there is no overall charge on the amino acid in its aqueous solution. Each amino acid has its own unique pI.

Q.50 [Long] How are amino acids separated by electrophoresis?

Electrophoresis is used to separate amino acids in a mixture on the basis of their charge. A few drops of the mixture are applied to a filter paper or gel placed in a buffered solution between two electrodes. When an electric field is applied, the amino acids separate according to their different pI values:

- If the pI of an amino acid is **greater** than the pH of the solution, it exists mainly in a form with a positive charge and migrates towards the **cathode**.
- If the pI is **lower** than the pH of the solution, it exists mainly in a form with a negative charge and migrates towards the **anode**.
- The greater the difference between pI and pH, the faster it migrates.
- If two amino acids have very similar pI values (such as glycine and leucine), the one with the larger molecular weight moves more slowly, because the charge has to carry a greater mass.

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Separation of amino acids at pH 6 by electrophoresis

Q.51 [Short] Why is ninhydrin used in electrophoresis?

Amino acids are colourless, so a detection technique is needed to see the location of the spots. **Ninhydrin** is used as a locating agent; it reacts with amino acids to produce a purple product (with H_2O, CO_2 and RCHO as by-products).

Q.52 [Short] What is the limitation of electrophoresis?

Electrophoresis cannot be used to separate large quantities of amino acids. It is only an analytical method for determining the number of amino acids in a mixture. To actually separate an entire mixture, other laboratory techniques are used, such as column chromatography.

Q.53 [Short] At pH 6, towards which electrode will lysine (pI = 9.74) and glutamic acid (pI = 3.22) move? What about alanine (pI = 6.02)?

The pI of lysine is greater than the pH, so it is positively charged and moves towards the cathode. The pI of glutamic acid is lower than the pH, so it is negatively charged and moves towards the anode. The pI of alanine is almost equal to the pH, so it has no overall charge and stays near the starting point.

Q.54 [Long] What are peptides? How is a peptide bond formed?

Amino acid molecules can react with each other: the acidic –COOH group of one molecule reacts with the basic $–NH_2$ group of another. When two amino acids react together, the resulting molecule is called a **dipeptide**. The amide bond, –CO–NH–, between two amino acid molecules is called a **peptide bond** or peptide link. The reaction is a condensation reaction, because a small molecule, water, is eliminated when the reactant molecules join.

$$H_2N–CH(R)–COOH + H_2N–CH(R)–COOH \rightarrow H_2N–CH(R)–CO–NH–CH(R)–COOH + H_2O$$

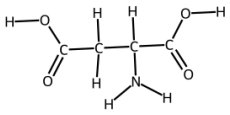
Q.55 [Short] Define tripeptide, polypeptide and isoelectric point.

Term	Meaning
Tripeptide	A compound formed from the condensation reaction between three amino acids.
Polypeptide	A compound formed from the condensation reaction between a large number of amino acids.
Isoelectric point	The pH value at which there is no overall charge on a particular amino acid in its aqueous solution.

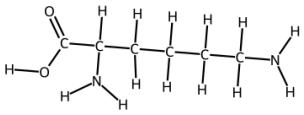
Quick Check 24.5

Q.56 [Short] The R groups in the amino acids aspartic acid and lysine are ($-CH_2-COOH$) and ($-CH_2-(CH_2)_3-NH_2$) respectively. a) Draw the displayed formulae of these amino acid. b) Draw the structures of their Zwitter ions. c) Draw the structures of their ions in acidic conditions. d) Draw the structures of their ions in basic conditions. Discuss with what we mean by the isoelectric point of these amino acids.

a) Displayed formulae:



aspartic acid



lysine

	Aspartic acid	Lysine
Formula	$H_2N–CH(CH_2–COOH)–COOH$	$H_2N–CH[(CH_2)_4–NH_2]–COOH$
b) Zwitterion	$H_3N^+–CH(CH_2–COOH)–COO^-$	$H_3N^+–CH[(CH_2)_4–NH_2]–COO^-$
c) Acidic conditions	$H_3N^+–CH(CH_2–COOH)–COOH$	$H_3N^+–CH[(CH_2)_4–NH_3^+]–COOH$
d) Basic conditions	$H_2N–CH(CH_2–COO^-)–COO^-$	$H_2N–CH[(CH_2)_4–NH_2]–COO^-$

Isoelectric point: it is the pH at which the amino acid has no overall charge in aqueous solution, i.e. the concentration of its zwitterion is at a maximum, so it does not move in an electric field. Each amino acid has its own pI. Aspartic acid has an extra acidic –COOH group, so its isoelectric point lies at a low (acidic) pH; lysine has an extra basic $–NH_2$ group, so its isoelectric point lies at a high (basic) pH (9.74).

Zwitterions are shown in the general textbook form; the low pI of aspartic acid is general knowledge (the book gives the pI of lysine only).